Nathan Chan
Metallurgical Laboratory

Summer 2026 Co-op – Final Presentation

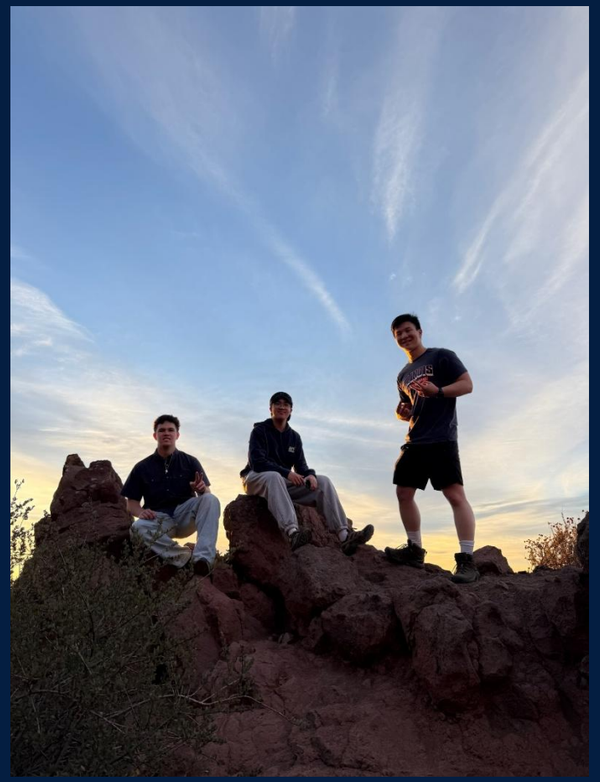


Safety First	Act with Integrity	Empower our People	Be Accountable
Play to Win	Drive Innovation	Quality Focused	

Introduction

- University of Illinois Urbana-Champaign
  - Sophomore
  - Major: Materials Science & Engineering
  - Minor: Hoeft Technology & Management
  - Graduation: May 2029
  - Hometown: Pasadena, CA
- Experiences:
  - Current: Co-op at Dana
  - Previous: California Institute of Technology
- Hobbies/Interests:
  - Water Polo
  - Hiking/Camping









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My Role & Goals

- Goals
  - Apply material science in automotive industry
  - Build meaningful, lasting relationships
  - Learn material and failure analysis technique and knowledge

- Accomplishments
  - 18 Completed Service Requests, 3 in Review
  - Knowledge in Heat Treatment, Carburization, Castings, Coatings, Welds, Part Nomenclature, & Tools/Techniques





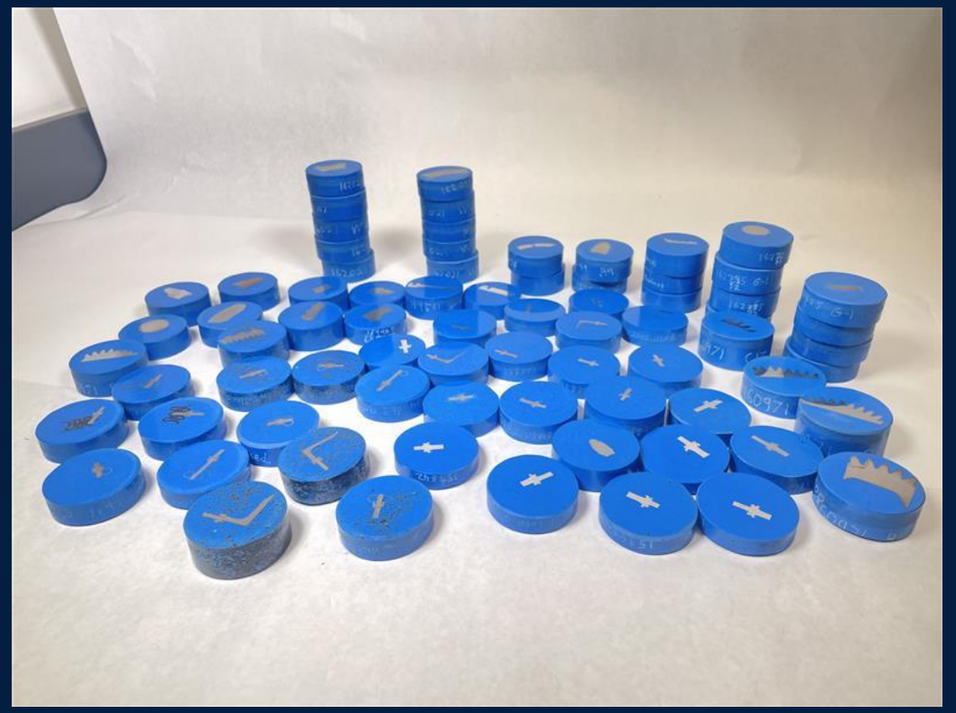


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Material Characterization

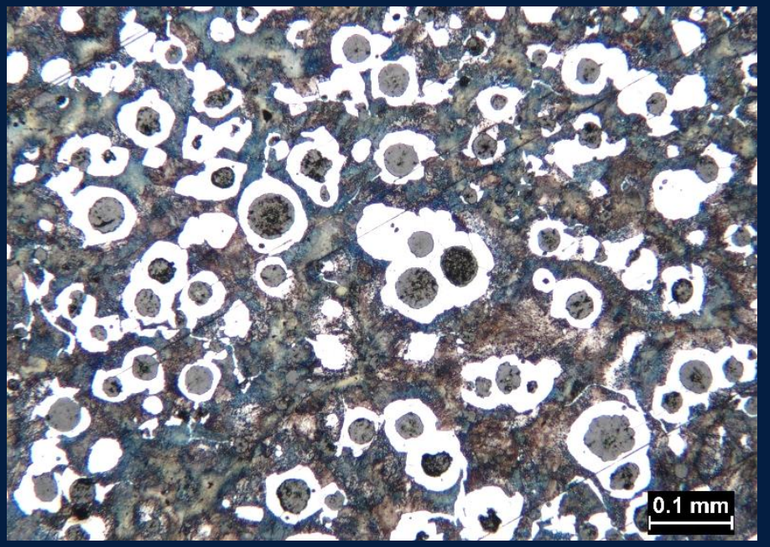
- Sample preparation
- Hardness testing

  (Brinell, Rockwell, Vickers)
- Chemistry verification
- Microstructure evaluation





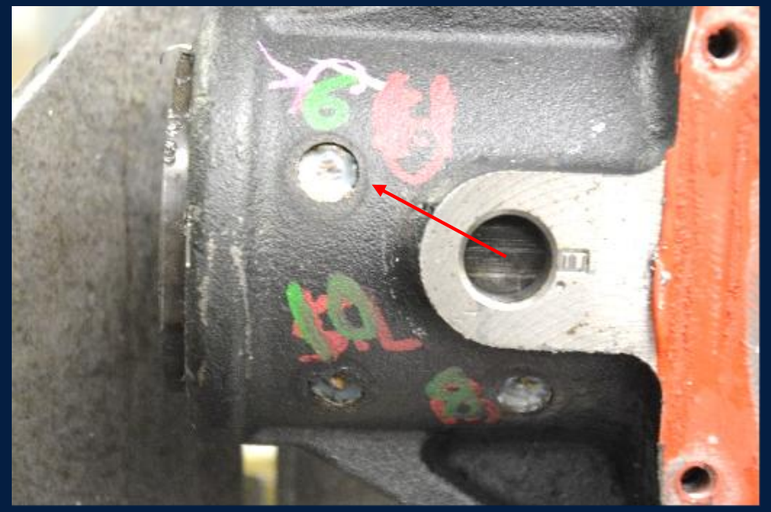


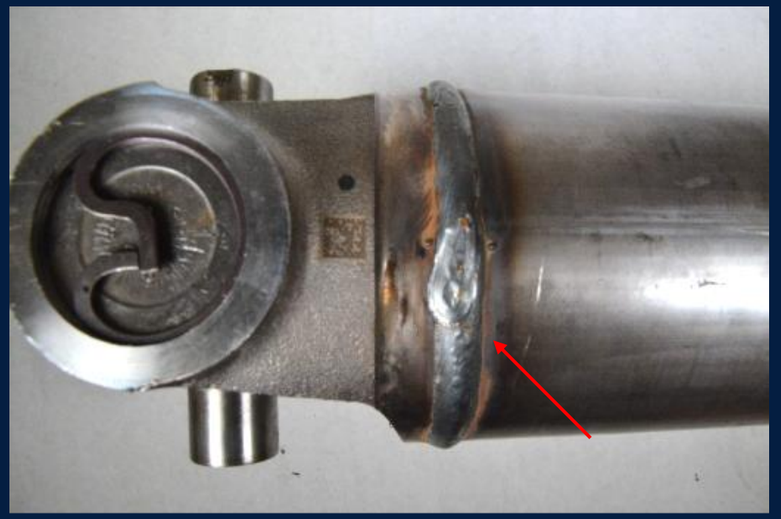


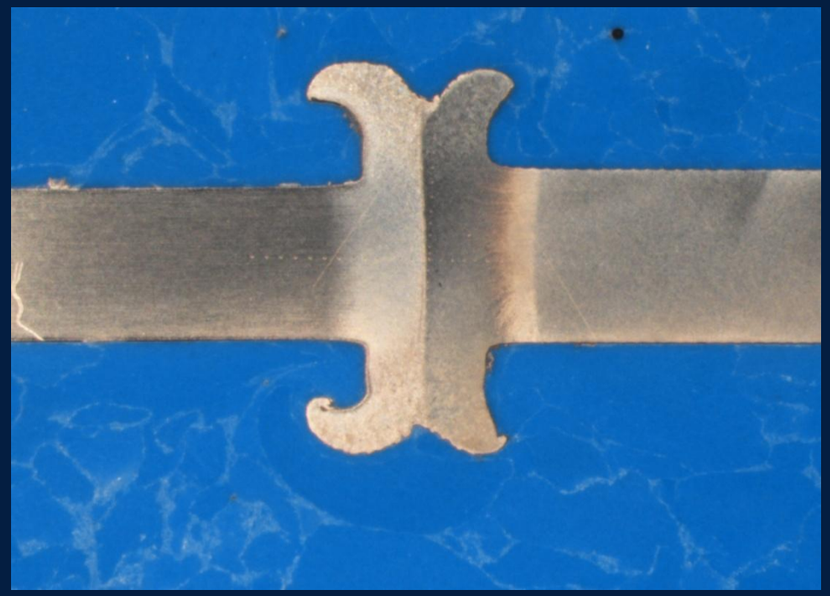
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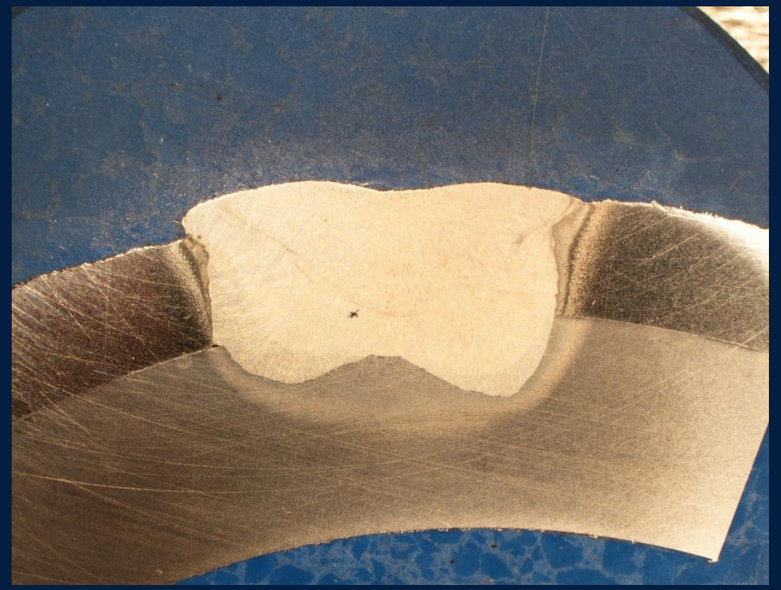
Weld Geometry

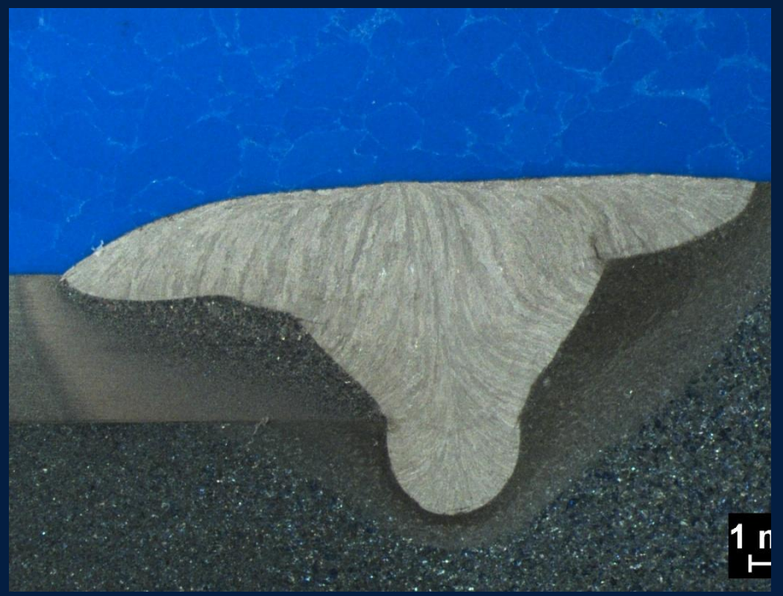












Friction Weld

Puddle Weld

Circle Weld

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  - Dana Combs, Lisa Hendrzak, Anthony Archer
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- Hiring Team
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